# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Seventh Semester of B. Tech. Examination (CL)**

**November 2013**

**CL403 Design of Structures-I**

**Date: 19.11.2013, Tuesday** **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

*Instructions:*

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 456:2000, IS 800:2007, IS 875, SP 16, and Steel Table are permitted.

## SECTION – I

Q - 1 Design and detail a beam of 6 m effective span, carrying a uniform load of 25 kN/m, if the compression flange is laterally unsupported. Assume $f_y$ = 250 MPa. Give any two checks. [07]

Q - 2 (a) Design and detail a double angle tension member connected on each side of a 10 mm thick gusset plate, to carry an axial factored load of 350 kN. Use welded connection. The length of the member is 3 m. [07]

(b) Design a built-up column consisting of two channels placed back-to-back. The column carries an axial compressive factored load of 1500 kN. The effective height of the column is 5 m. Also design and detail the lacing system. Use bolted connection. [07]

**OR**

Q - 2 (a) A tension member of truss consists of a single angle ISA 125 x 75 x 10 mm is carrying a factored load of 280 kN. 20 mm diameter bolts are used for the connection. Design and detail the connection to a gusset plate using a lug angle. [07]

(b) Design and detail a double angle strut to carry an axial compressive factored load of 250 kN. The length of the strut is 3.0 m. Welded connections are to be used to connect the strut to 12 mm thick gusset plate. [07]

Q - 3 **Answer the following questions:**

1) Enlist two advantages of HSFG bolts over Bearing type bolts. [02]
2) Enlist two advantages of bolted connections over welded connections. [02]
3) State any two assumptions used in design of bearing bolts. [02]
4) Explain the following terms used in bolted connections and show by means of sketches: [04]
   Gauge distance, Tacking bolts
5) Explain following types of welded connections by means of sketches: [04]
   Fillet weld & Butt weld

**OR**

Q - 3 **Answer the following questions:**

1) Enlist two advantages of welded connections over bolted connections. **[02]**
2) Explain following types of bolted connections by means of sketches: **[04]**
   Lap Joint & Butt Joint
3) Explain the following terms used in bolted connections and show by means of sketches: **[04]**
   Pitch of the bolts, Edge distance
4) Explain following types of welded connections by means of sketches: **[04]**
   Slot weld & Plug weld

## SECTION – II

Q - 4 **Answer the following questions:** **[07]**

1. Limit state method disallows over-reinforced section. Why?
2. What is the value of design bond stress for concrete grade M25 and HYSD bars in compression?
3. Minimum grade of concrete to be used for RCC work is __________.
4. Why minimum shear reinforcement is required?
5. Since concrete is strong in compression then why reinforcement is required?
6. State IS specifications for pitch and diameter of lateral ties.
7. Why dowels are provided in footings?

Q - 5 (a) A R.C. beam 300 mm × 450 mm in cross-section is reinforced with 3 nos. 20 mm diameter bars of grade Fe 250, with an effective cover of 50 mm. The ultimate shear at the section is 138 kN. Design and detail the shear reinforcement if 1 bar of 20 mm diameter is bent at 45° to resist shear. Use M20 concrete. **[07]**

(b) A R.C. column 450 mm × 500 mm in cross section has an unsupported length of 3.75 m. The column is effectively held in position and restrained against rotation at one end and at the other end restrained against rotation but not held in position. Check whether the column is short or long. **[02]**

(c) Design and detail the reinforcement in spiral column of 400 mm Ø subjected to factored load of 1500 kN. A column has a unsupported length of 3.3 m and it is braced against side sway. Use M20 concrete and Fe 415 steel. **[05]**

**OR**

Q - 5 (a) Design with detailing of simply supported tee beam of span 7 m and spaced at 3 m centers. The thickness of slab is 100 mm and total characteristic load including self-weight of beam is 30 kN/m. The overall size of the beam is 230 mm × 600 mm. Check this beam for deflection and cracking only. Use M20 concrete and Fe 415 steel. **[07]**

(b) Design using charts, the reinforcement of grade Fe 415 steel for a R.C. column 230 mm wide and 400 mm deep to resist an ultimate axial load of 1270 kN and an ultimate bending moment of 58 kN-m about major axis bisecting depth of the column. Unsupported length of column is 4.5 m, and effective length of column is 2.75 m. Use M20 concrete. **[07]**

Q - 6 (a) A slab is simply supported over a clear opening in plan 6 m × 4.5 m. The corners of the slab are free to lift up. There is a 50 mm thick plain concrete floor finish on the top of the slab. Live load is 3 kN/m$^2$. Design and detail this slab. Also check this slab for control of deflection, shear and cracking. Use M20 concrete and Fe 415 steel. **[09]**

(b) An isolated footing for an R.C. column of size 230 mm x 230 mm carries a vertical load of 500 kN. The safe bearing capacity of soil is 200 $kN/m^2$. Design and detail the footing for flexure only. Use M20 concrete and Fe 415 steel. [05]

**OR**

Q - 6 (a) Design and show detailing of a simply supported slab for a room 6 m × 2.77 m provided over a masonry wall of 230 mm thick. Live load is 2 $kN/m^2$ and floor finish of 1 $kN/m^2$. Check this slab for control of deflection, shear and cracking. Use M20 concrete and Fe 415 steel. [09]

(b) A square footing 3 m × 3 m and 600 mm thickness supports a 450 mm × 450 mm column. Design the dowels between the column and the footings if the characteristic loads at the base of column are 780 kN live load and 1000 kN dead load. Use M20 concrete and Fe 415 steel. [05]

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